

CONCEPT 30.2

Gymnosperms bear “naked” seeds, typically on cones



Nonvascular plants (bryophytes)
Seedless vascular plants
Gymnosperms
Angiosperms

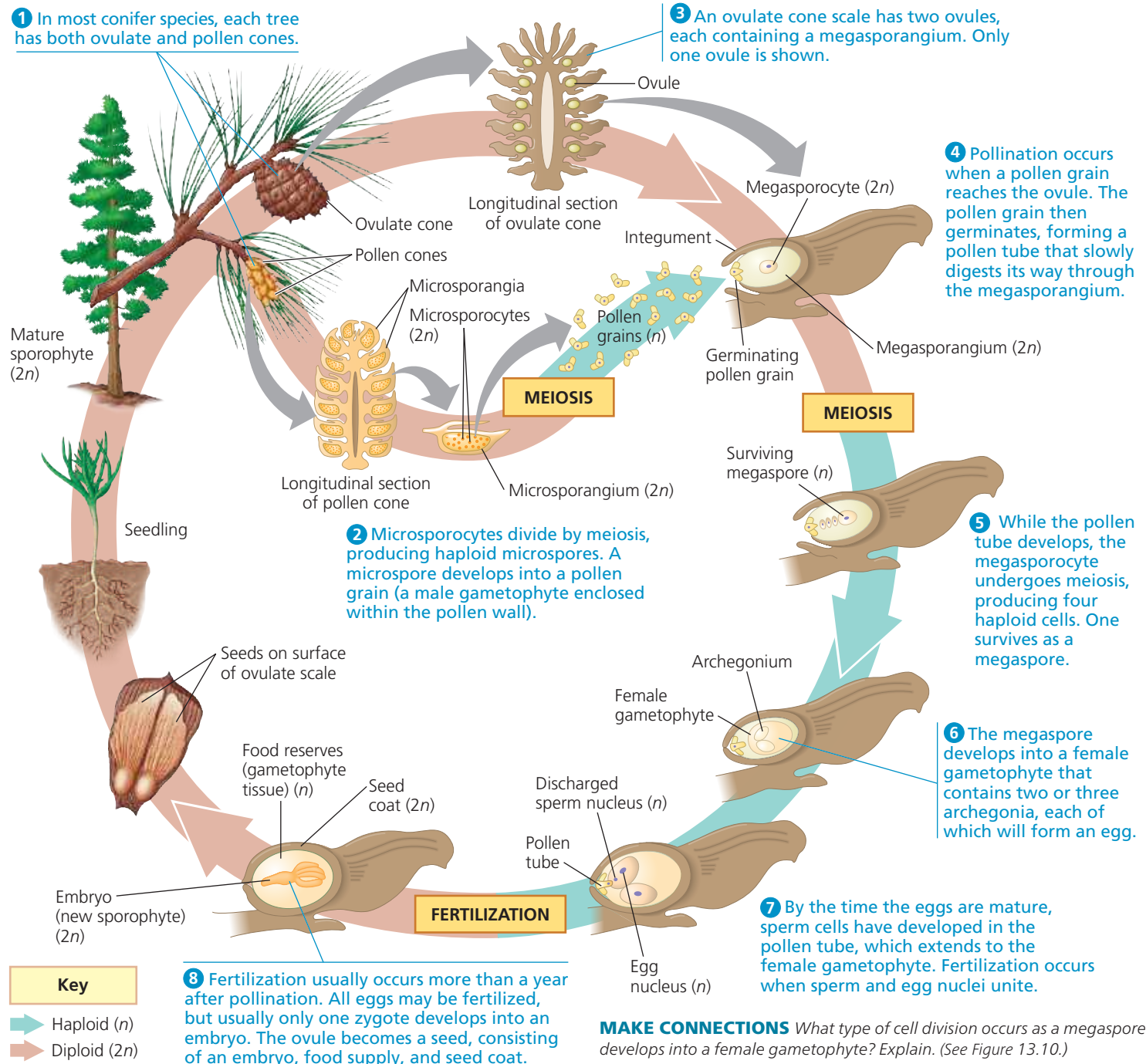
Extant seed plants form two sister clades: gymnosperms and angiosperms. Recall that gymnosperms have “naked” seeds exposed on sporophylls that usually form cones. (Angiosperm seeds are enclosed in chambers that mature

into fruits.) Most gymnosperms are cone-bearing plants called **conifers**, such as pines, firs, and redwoods.

The Life Cycle of a Pine

As you read earlier, seed plant evolution has included three key reproductive adaptations: the miniaturization of their gametophytes; the advent of the seed as a resistant, dispersible stage in the life cycle; and the appearance of pollen as an airborne agent that brings gametes together. **Figure 30.4** shows how these adaptations come into play during the life cycle of a pine, a familiar conifer.

▼ **Figure 30.4** The life cycle of a pine.



MAKE CONNECTIONS What type of cell division occurs as a megaspore develops into a female gametophyte? Explain. (See Figure 13.10.)

➔ **Mastering Biology Animation: Pine Life Cycle**